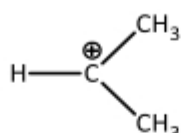


11 Alkenes

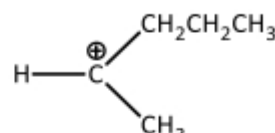
Section A

1 (a)



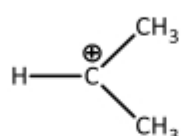
The presence of more electron-donating alkyl groups stabilises the carbocation by reducing its positive charge.

(b)



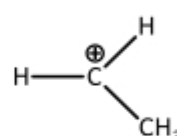
The presence of more electron-donating alkyl groups stabilises the carbocation by reducing its positive charge.

(c)



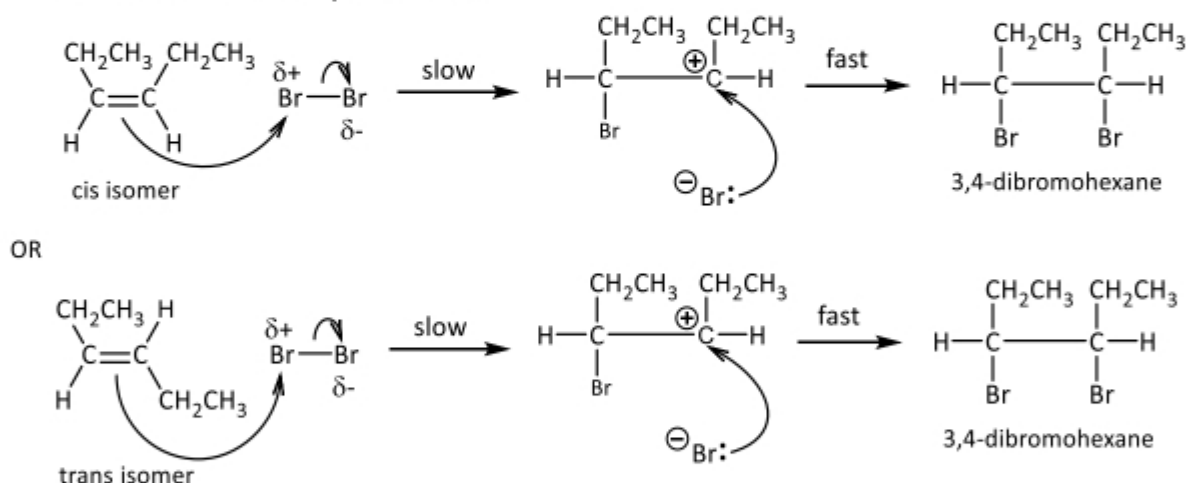
The presence of electron-withdrawing chlorine destabilises the other carbocation by intensifying its positive charge.

(d)



The presence of electron-withdrawing fluorine destabilises the other carbocation even though it is one carbon away.

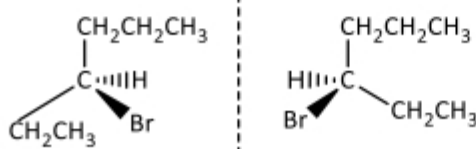
2 (a) Name of mechanism: Electrophilic addition



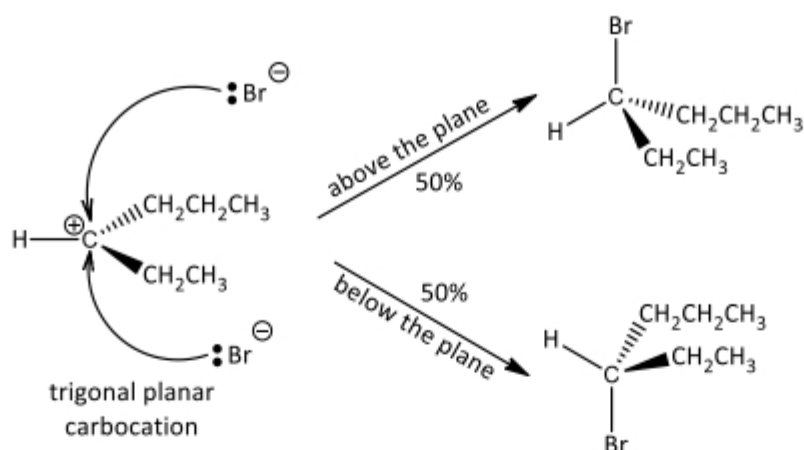
Things to take note when asked to describe mechanism for electrophilic addition

- ✓ Indicate "electrophilic addition" as name of mechanism
- ✓ Correct structures (ambiguous formula NOT acceptable)
- Slow step:
 - ✓ Indicate "slow" for first step (formation of carbocation)
 - ✓ Partial charges on electrophile Br_2
 - ✓ 2 arrows for slow step (one from $\text{C}=\text{C}$ bond to Br, the other from $\text{Br}-\text{Br}$ bond to Br)
- Fast step:
 - ✓ Lone pair on nucleophile Br^-
 - ✓ 1 arrow for fast step from lone pair on Br^- to C (not to +)

(b) (i)

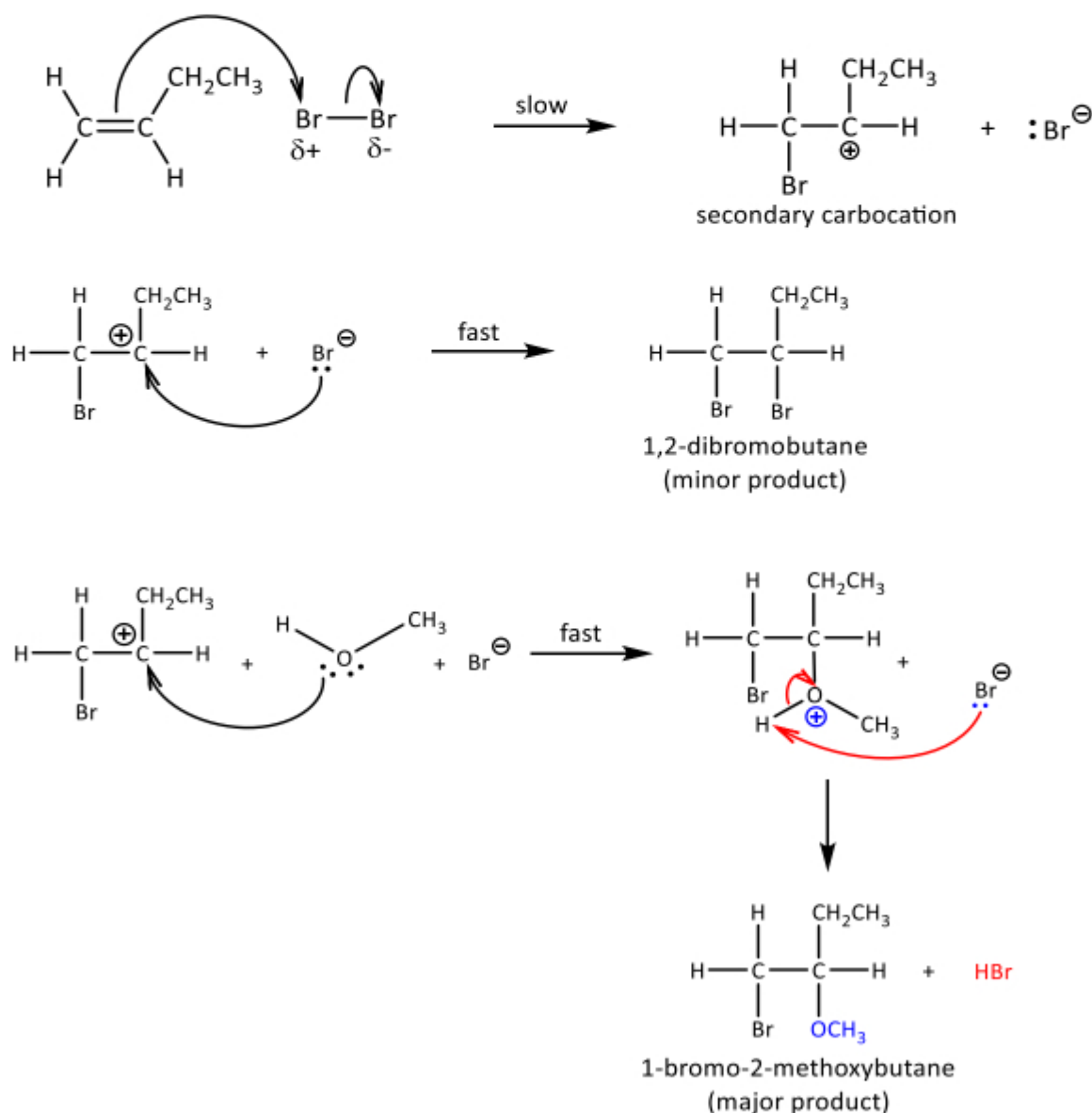


(ii)



The carbocation is planar with respect to the electron deficient carbon, and has equal chances of being attacked from the top or bottom by the bromide ion in the second step of the reaction. Therefore, there are equal chances of producing the 2 enantiomers.

3



2-bromo-1-methoxybutane is **not** formed hence the products are formed from the secondary carbocation. The secondary carbocation is more likely to be formed than the primary carbocation because it is more stable than the primary carbocation due to presence of two electron-donating alkyl groups on positive carbon than one that stabilises the carbocation by dispersing the positive charge.

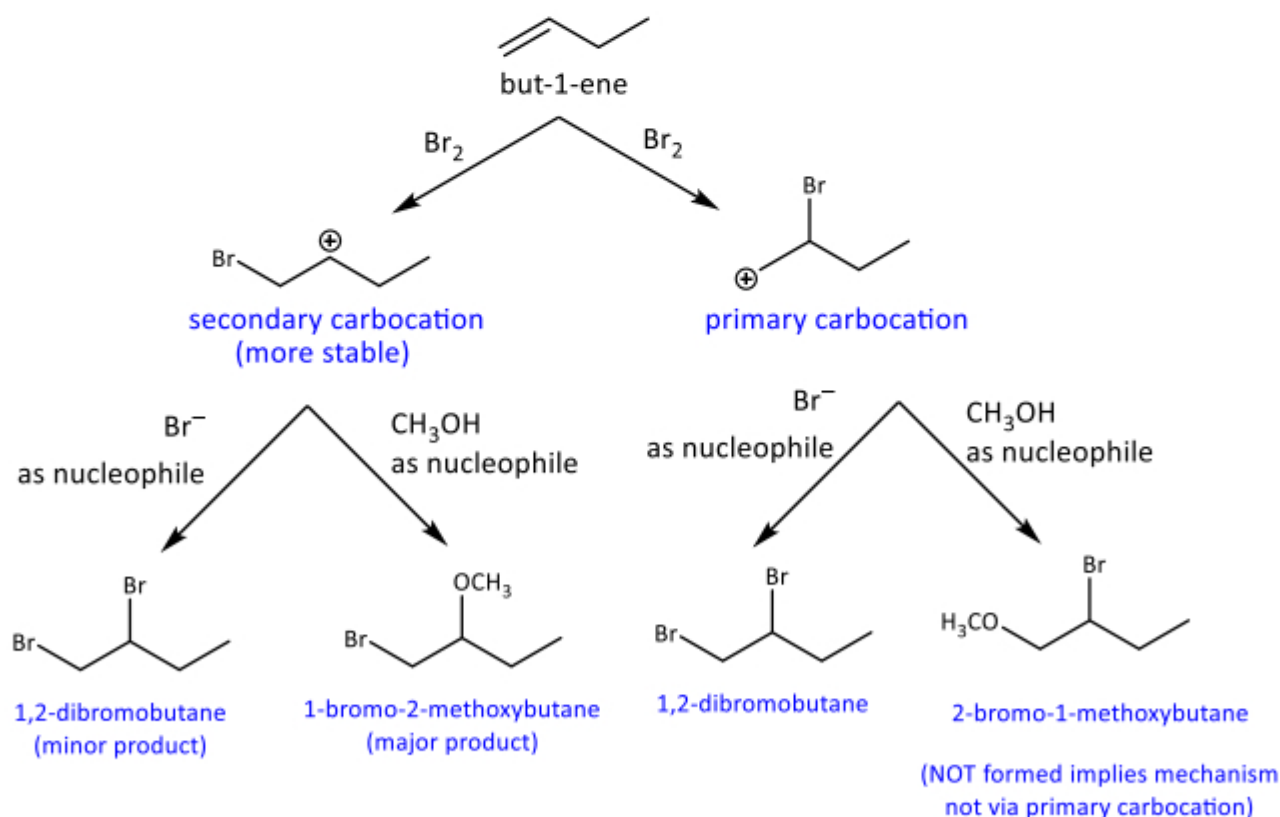


Since both nucleophiles CH_3OH and Br^- can attack the secondary carbocation intermediate, a mixture of 1-bromo-2-methoxybutane and 1,2-dibromobutane will be obtained.

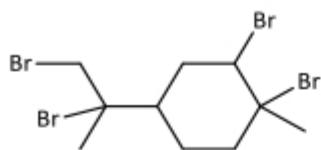
1-bromo-2-methoxybutane is the major product because methanol (as a solvent) is present in a much higher concentration compared to the bromide ion and therefore attacks the intermediate preferentially in the second step of the reaction.

Note:

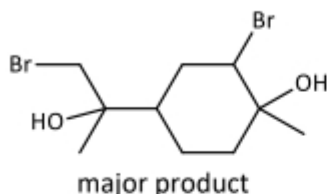
- In the first step, the electron-rich alkene attacks the electrophile Br_2 , forming a more stable carbocation intermediate with the positive charge on the carbon with more electron-donating alkyl groups.
- In the second step, there are two possible electron-rich species (CH_3OH and Br^-) that can attack the carbocation, as explained in the answers above.



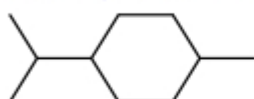
- 4 (a) Observation: reddish-brown liquid Br_2 decolourised (Note: you can refer to Data Booklet for the colours of Br_2 in different states)



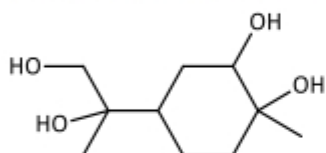
- (b) Observation: yellow-orange aqueous Br_2 decolourised



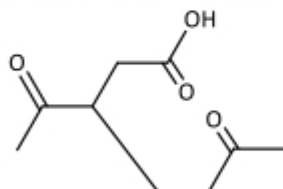
- (c) Observation: pressure gauge needle indicates lowered pressure after a while (Note: because after reaction, there is less hydrogen gas)



- (d) Observation: purple KMnO_4 decolourised and brown ppt of MnO_2 formed (Note: this is mild oxidation and MnO_2 is formed when there is alkaline medium)

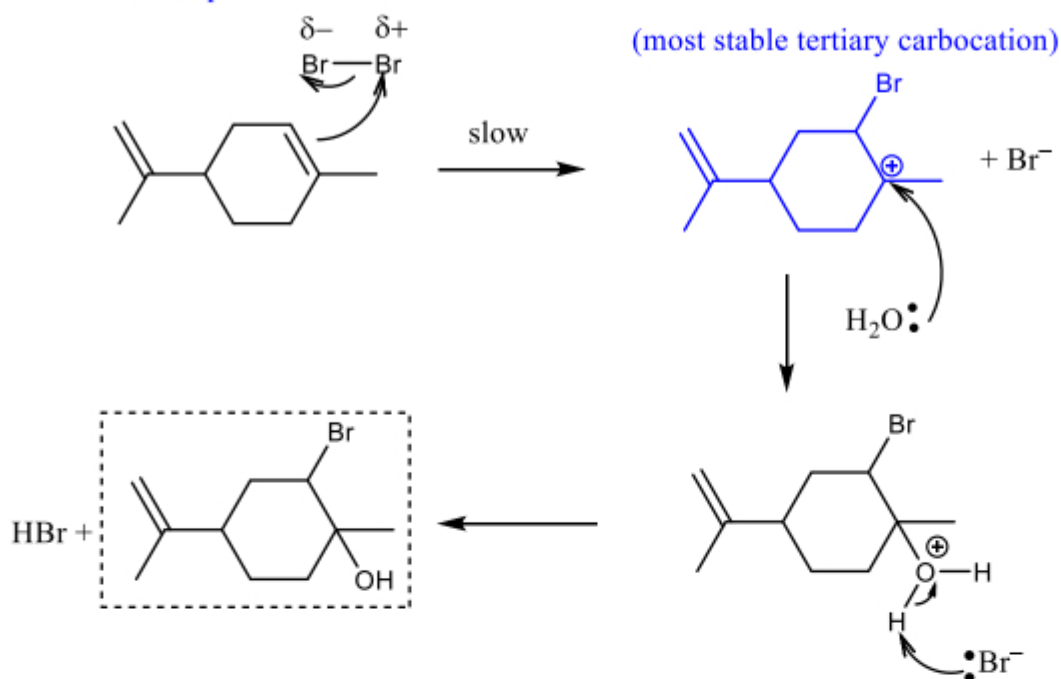


- (e) Observation: purple KMnO_4 decolourised, effervescence of (colourless and odourless) CO_2 gas (which gave a white precipitate of CaCO_3 when bubbled in limewater $\text{Ca}(\text{OH})_2$) (Note: CO_2 is formed from the terminal $=\text{CH}_2$ on the left)

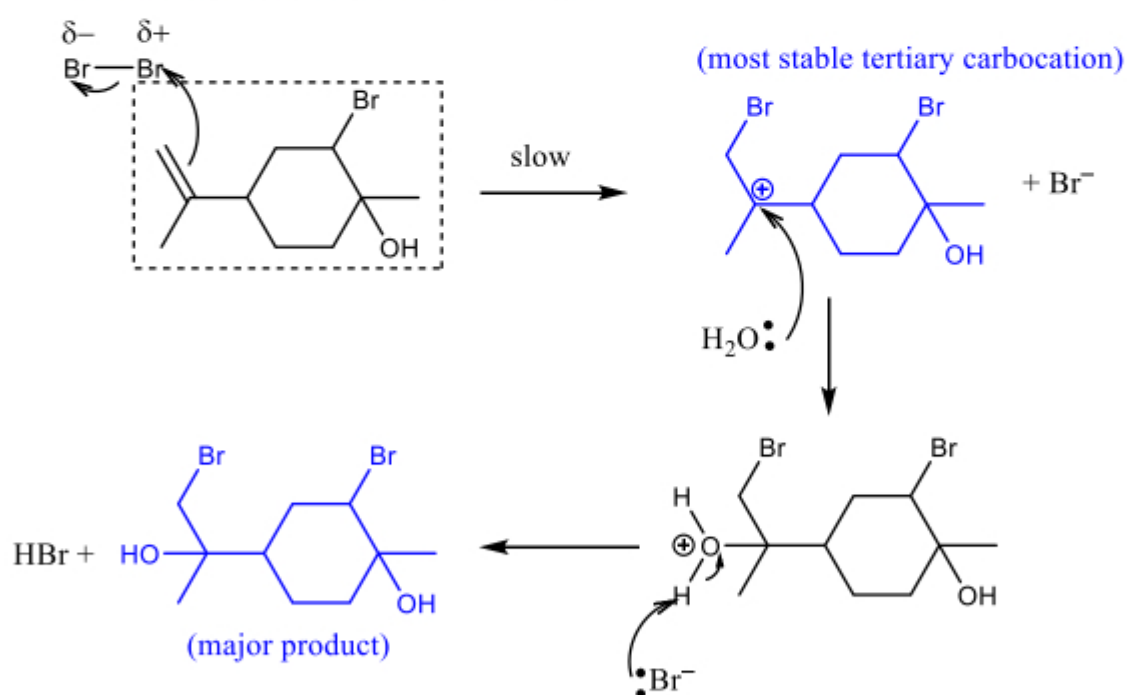


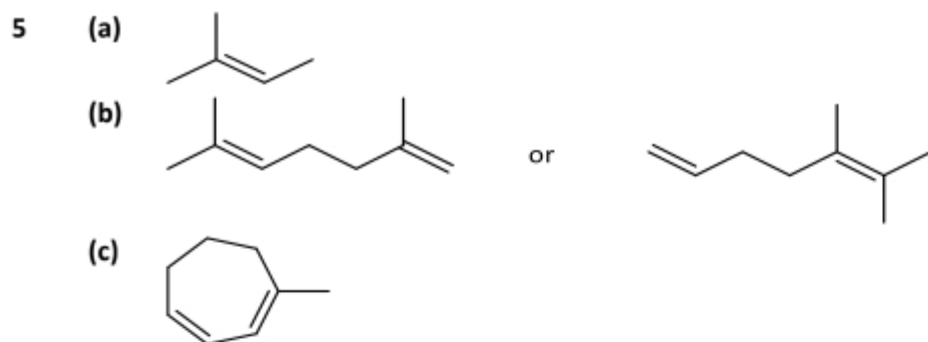
Teaching solution for Q4b:

First electrophilic addition mechanism



Second electrophilic addition mechanism

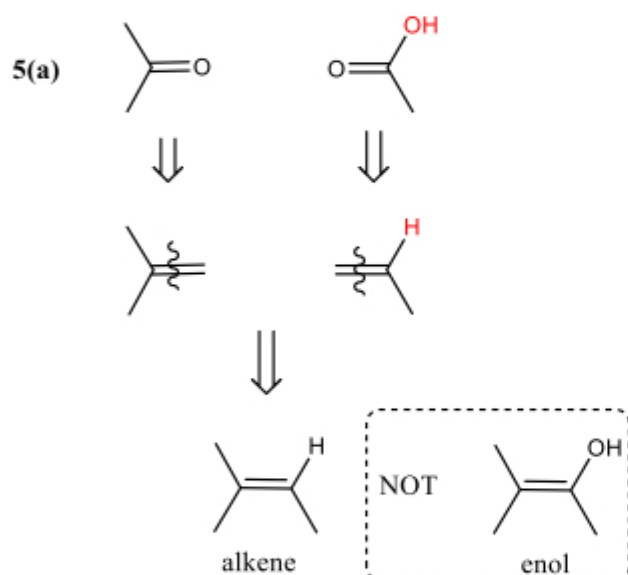




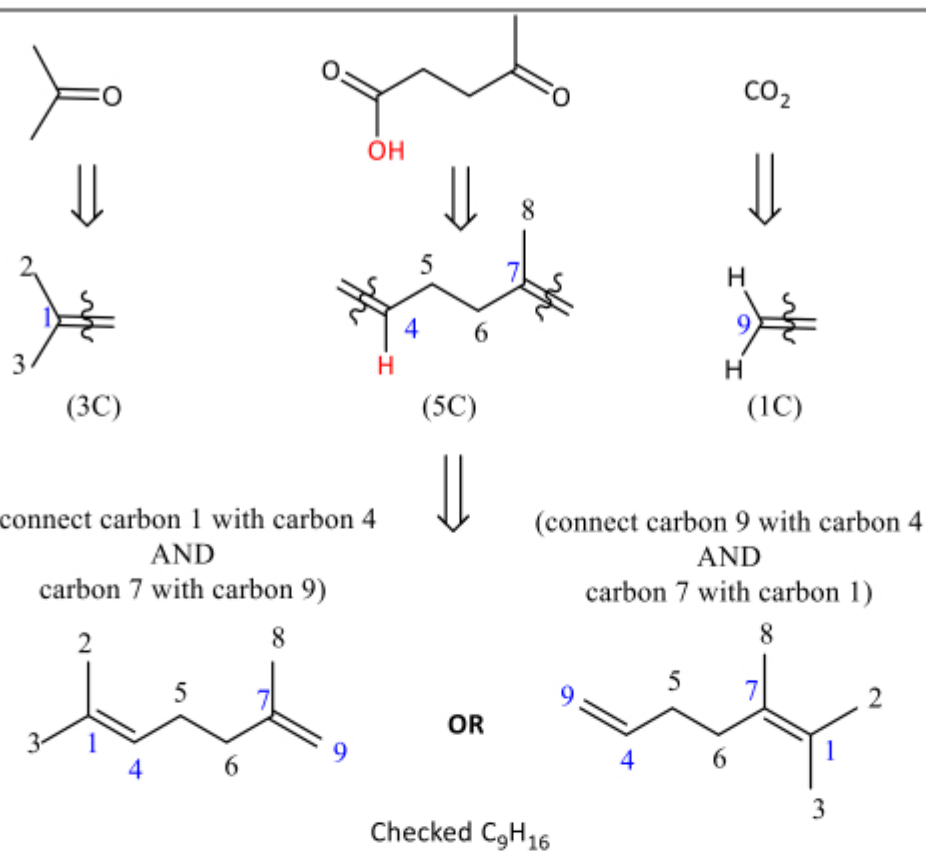
Note: count the number of carbons to ensure the number of carbons before and after oxidative cleavage are the same. For (c), you are expected to infer that 2 moles of CO_2 gas are given off in order to derive an alkene with C_8H_{12} , therefore it must contain a $=\text{CH}-\text{CH}=$ fragment (refer to notes on oxidative cleavage).

Teaching solution

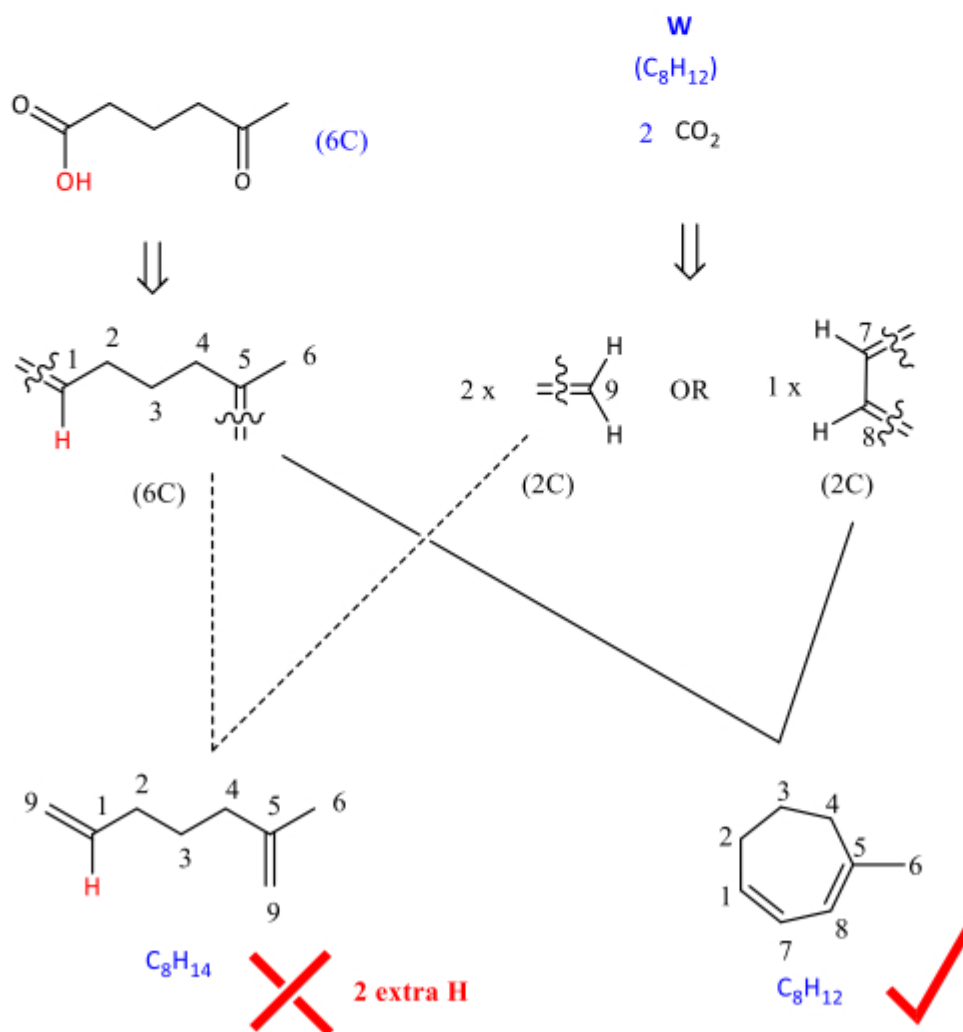
The following is known as retrosynthesis (you think backwards starting from products to find out what are the starting materials)



5(b)



5(c)



6 (a) Reagents and conditions: Br_2 in CCl_4 , absence of UV light (to prevent free radical substitution)

Observations:

- cyclohexane: reddish brown Br_2 not decolourised
- cyclohexene: reddish brown Br_2 decolourised

OR Reagents and conditions: aqueous Br_2

Observations:

- cyclohexane: yellow-orange $\text{Br}_2(\text{aq})$ not decolourised
- cyclohexene: yellow-orange $\text{Br}_2(\text{aq})$ decolourised

OR Reagents and conditions: KMnO_4 , dilute H_2SO_4 , heat

Observations:

- cyclohexane: Purple KMnO_4 not decolourised
- cyclohexene: Purple KMnO_4 decolourised

OR Reagents and conditions: KMnO_4 , $\text{NaOH}(\text{aq})$, cold

Observations:

- cyclohexane: Purple KMnO_4 not decolourised
- cyclohexene: Purple KMnO_4 decolourised and brown ppt of MnO_2 formed

(b) Reagents and conditions: KMnO_4 , dilute H_2SO_4 , heat

Observations:

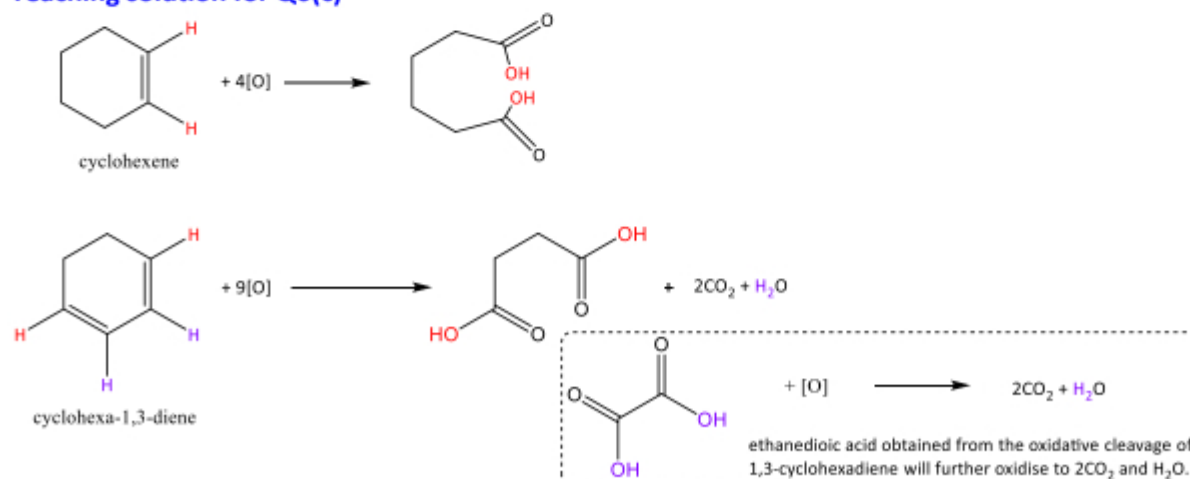
- but-1-ene: purple KMnO_4 decolourised, effervescence of (colourless and odourless) CO_2 gas (which gave a white precipitate of CaCO_3 when bubbled in limewater $\text{Ca}(\text{OH})_2$)
- but-2-ene: purple KMnO_4 decolourised but no gas evolved

(c) Reagents and conditions: KMnO_4 , dilute H_2SO_4 , heat

Observations:

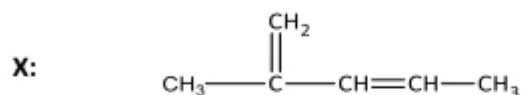
- cyclohexene: purple KMnO_4 decolourised but no gas evolved
- 1,3-cyclohexadiene: purple KMnO_4 decolourised, effervescence of (colourless and odourless) CO_2 gas (which gave a white precipitate of CaCO_3 when bubbled in limewater $\text{Ca}(\text{OH})_2$)

Teaching solution for Q6(c)

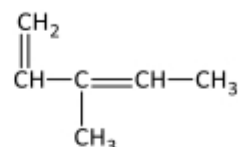


7 Deductions:

- **X** reacts with liquid bromine in the dark to form $C_6H_{10}Br_4 \Rightarrow$ **X** has 2 C=C double bonds and undergoes electrophilic addition reaction with liquid bromine
- $C_6H_{10} + 2Br_2 \rightarrow C_6H_{10}Br_4$
- Upon heating with acidified $KMnO_4$ solution, carbon dioxide gas was formed $\Rightarrow$ **X** is a terminal alkene; **X** undergoes oxidative cleavage
- $C_6H_{10} + 8[O] \rightarrow CH_3CO_2H + CH_3COCO_2H + CO_2 + H_2O$



2-methylpenta-1,3-diene

OR

3-methylpenta-1,3-diene

Note: In questions involving structural elucidation, you are required to explain qualitatively how you deduced the structures, by stating the type of reaction (eg. electrophilic addition, oxidative cleavage etc.), and type (and number) of functional groups responsible.

Teaching solution for Q7

